1-Loop Corrections in L-K Form

Reconciling corrections w. Spectral Density

We know that the Exact Propagator can be written in Lehmann-Källén form as:

$$\tilde{\Delta}(k^2) = \frac{1}{k^2+m^2-i\epsilon} + \int_{4m^2}^{\infty} ds\, \rho(s) \frac{1}{k^2+s-i\epsilon}$$

→ Spectral density (real and Non-negative)

We also know that to $O(g^2)$ in $d=6$ φ^3 theory:

$$\tilde{\Delta}(k^2) = \frac{1}{k^2+m^2-i\epsilon-\Pi(k^2)} \quad \text{where} \quad \Pi(k^2) = \frac{1}{2}\alpha\int_0^1 dx\, D \ln\left(\frac{D}{D_0}\right) - \frac{1}{2}\alpha(k^2+m^2) + O(\alpha^2)$$

Where all the parameters are defined in the notes for the one-loop corrections.

We know that we can rationalize complex numbers such that:

$$\frac{1}{x-i\epsilon} = \frac{x}{x^2+\epsilon^2} + \frac{i\epsilon}{x^2+\epsilon^2} = \underbrace{P\times\frac{1}{x}}_{\text{Principal Part}} + i\pi\delta(x)$$

Q: What is the principal part? Where did this $\delta(x)$ come from?

A: If we have: $\frac{x}{x^2+\epsilon^2} + \frac{i\epsilon}{x^2+\epsilon^2} = P.\frac{1}{x} + \frac{i\epsilon}{x^2+\epsilon^2}$

The principal part is the Cauchy principle value in complex analysis.

Due to ϵ being a very small increment, we can write: $\frac{i\epsilon}{x^2+\epsilon^2} = \underbrace{i\pi\delta(x)}_{\text{Normalization Factor}}$ $\epsilon \ll 1$ (Put it into graph to verify)

This gives us the imaginary part of the Lehmann-Kallen form as:

$$\text{Im}\,\tilde{\Delta}(k^2) = \pi\delta(k^2+m^2) + \int_{4m^2}^{\infty} ds\, \rho(s)\, \pi\delta(k^2+s)$$

$$= \pi\delta(k^2+m^2) + \pi\rho(-k^2)$$

Now, If $\text{Im}\,\tilde{\Delta}(k^2) = \pi\delta(k^2+m^2) + \pi\rho(-k^2)$, If $-k^2 = m^2$ then $\rho(-k^2) = 0$ $\because \rho(s) = 0$ for $s \geq 4m^2$. Therefore this imaginary part can be split as $\text{Im}\,\tilde{\Delta}(k^2) = \pi\delta(k^2+m^2)$ $k^2 \geq 4m^2$, $\pi\rho(-k^2)$ $-k^2 \geq 4m^2$

We can then write:

$$\pi\rho(s) = \text{Im}\,\tilde{\Delta}(-s) \quad \text{for } s \geq 4m^2$$

If we say that $\text{Im}\,\Pi(k^2) = 0$ for some range of k^2 (we saw that this is true for $k^2 > -4m^2$ at $O(\alpha)$) then we can get:

$$\text{Im}\,\tilde{\Delta}(k^2) = \pi\delta(k^2+m^2)\,\delta(\text{Re}\,\Pi(k^2)) \quad \text{Im}\,\Pi(k^2) = 0$$

$\mathrm{Im}\,\tilde{\Delta}(k^2) = \pi\delta(k^2+m^2-\Pi(k^2))$ for $\mathrm{Im}\,\Pi(k^2)=0$

$$\tilde{\Delta}(k^2) = \frac{1}{k^2+m^2-i\epsilon-\Pi(k^2)} = P\frac{1}{x} + i\pi\delta(k^2+m^2-\Pi(k^2))$$

This is counted as purely real.

Since $\Pi(-m^2)=0$, $\Pi'(-m^2)=0$ we can happily write:

$\mathrm{Im}\,\tilde{\Delta}(k^2) = \pi\delta(k^2+m^2)$ for $\mathrm{Im}\,\Pi(k^2)=0$

Notice now $\mathrm{Im}\,\Pi(k^2)=0$ corresponds to $f(-k^2)=0$

Now for $\mathrm{Im}\,\Pi(k^2)\neq 0$ for some range of k^2, then by ignoring the $i\epsilon$, we can write:

$$\mathrm{Im}\,\tilde{\Delta}(k^2) = \frac{\mathrm{Im}\,\Pi(k^2)}{(k^2+m^2+\mathrm{Re}\,\Pi(k^2))^2+(\mathrm{Im}\,\Pi(k^2))^2} \quad \text{for } \mathrm{Im}\,\Pi(k^2)\neq 0$$

We know from previous notes that $\mathrm{Im}\,\Pi(k^2)\neq 0$ for $k^2<-4m^2$. Therefore we can compare this condition to the one we found for the Lehmann-Kallen form.

$$\pi\rho(s) = \frac{\mathrm{Im}\,\Pi(-s)}{(-s+m^2+\mathrm{Re}\,\Pi(-s))^2+(\mathrm{Im}\,\Pi(-s))^2}$$

This agrees spectacularly for our independent analysis of $\Pi(k^2)$.

Another way of looking at this is:

$$\Pi(k^2) = \frac{1}{2}\alpha\int_0^1 dx\,\ln\left(\frac{D}{D_0}\right) - \frac{1}{12}\alpha(k^2+m^2) + O(\alpha^2) \quad ; \quad D = x(1-x)k^2+m^2-i\epsilon$$

↳ Real only as D/D_0 are positive

D is real and positive as long as $x(1-x)k^2 \geq m^2$. The maximum value of $x(1-x)$ is $1/4$.

$\therefore D$ is " as long as $k^2 \geq -4m^2$. This implies $\mathrm{Im}\,\Pi(k^2)=0$. For $x(1-x)k^2 < m^2$ we have $\mathrm{Im}\,\Pi(k^2)\neq 0$.